

AWS KMS & Secrets Management

Importance of KMS & Secrets Management

AWS KMS

- Protects sensitive data using strong encryption keys.
- Ensures compliance (ISO, HIPAA, PCI-DSS).
- Gives centralized control over who can encrypt/decrypt data.
- Prevents data exposure if storage is compromised.

AWS Secrets Manager

- Eliminates hardcoded passwords and API keys.
- Automatically rotates credentials without app downtime.
- Controls access using IAM (least privilege).
- Audits secret access via CloudTrail.

Tools Used in This Implementation

- AWS KMS – Key creation, rotation, access control
- Amazon S3 – Encrypted object storage
- AWS Secrets Manager – Secure credential storage
- IAM – Role and permission management
- CloudTrail – Logging and auditing access
- EC2 / Lambda – Application access testing

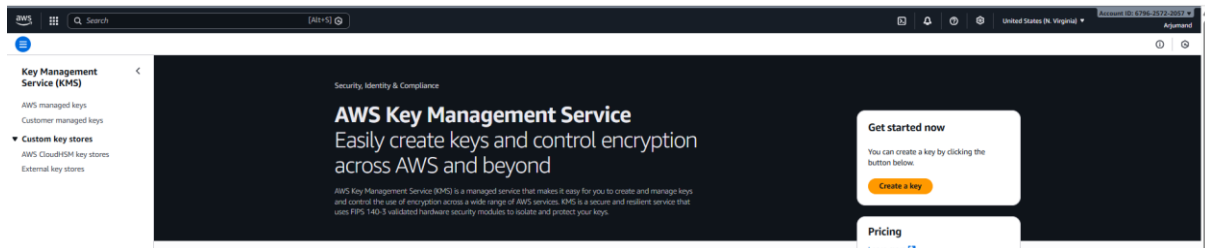
1. How do you encrypt an S3 bucket using a customer-managed KMS key?

Steps to Follow

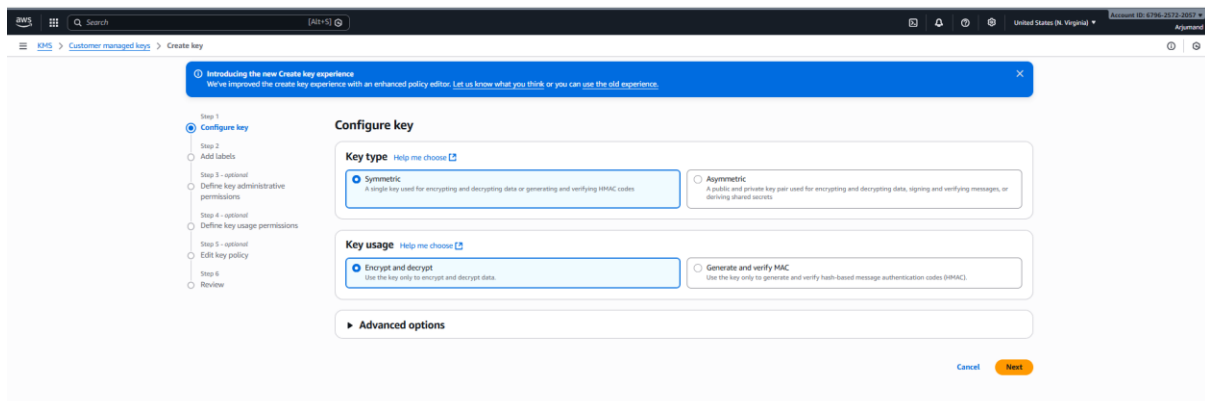
1. Create a symmetric Customer-Managed KMS key in AWS KMS.
2. Enable automatic key rotation for compliance.

3. Create or select an S3 bucket in the same region.
4. Enable default encryption on the bucket.
5. Choose SSE-KMS as encryption type.
6. Select the customer-managed KMS key (CMK).
7. Upload an object and verify it shows aws:kms encryption.

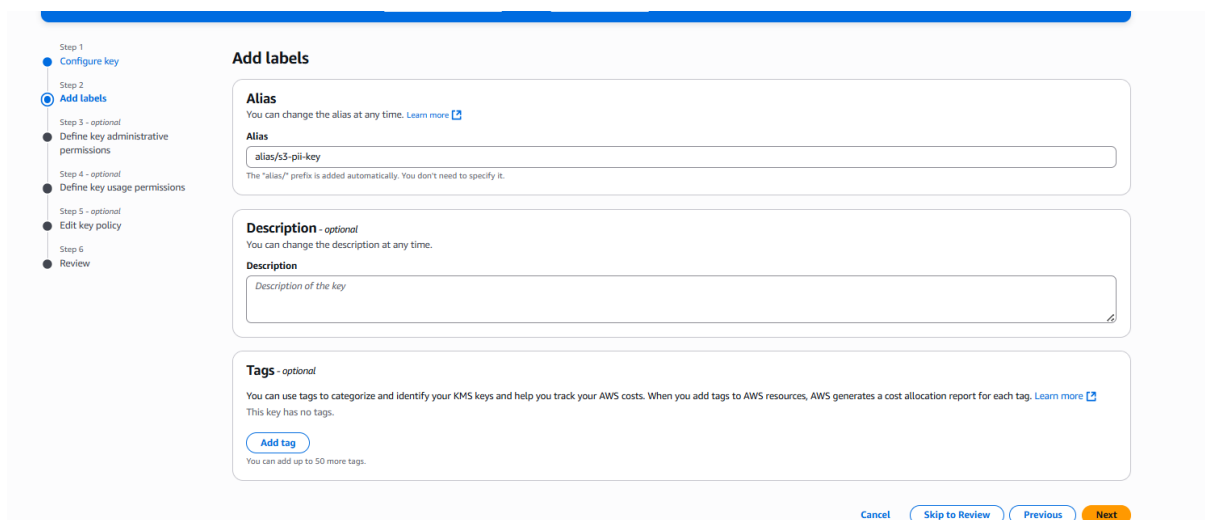
Go to AWS console and search for KMS and navigate to KMS



Select N.verginia region and click to create a key and choose option symmetric key, it will give same key for encryption and decryption.



Given key name



I have given permission to this IAM users

Step 1
Configure key

Step 2
Add labels

Step 3 - optional
Define key administrative permissions

Step 4 - optional
Define key usage permissions

Step 5 - optional
Edit key policy

Step 6
Review

Define key administrative permissions - *optional*

Key administrators (2/25)
Select the IAM users and roles authorized to manage this key via the KMS API. These administrators will be added to the key policy under the statement identifier (Sid) 'Allow administration of the key'. Modifying this Sid might impact the console's ability to update the administrator statement in the key policy. [Learn more](#)

Q Search Key administrators

☐	Name	Path	Type
☒	Admin	/	User
☒	ARJM	/	User
☐	Devops	/	User
☐	vpc-user	/	User
☐	AWSServiceRoleForAPIGateway	/aws-service-role/ops.apigateway.amazonaws.com/	Role
☐	AWSServiceRoleForAutoScaling	/aws-service-role/autoscaling.amazonaws.com/	Role
☐	AWSServiceRoleForEC2Spot	/aws-service-role/spot.amazonaws.com/	Role
☐	AWSServiceRoleForElasticLoadBalancing	/aws-service-role/elasticloadbalancing.amazonaws.com/	Role
☐	AWSServiceRoleForGlobalAccelerator	/aws-service-role/globalaccelerator.amazonaws.com/	Role
☐	AWSServiceRoleForNATGateway	/aws-service-role/ec2-nat-gateway.amazonaws.com/	Role

Key deletion
☒ Allow key administrators to delete this key.

Cancel Skip to Review Previous Next

I have kept policy as it is as of now.

Key policy

Review the key policy statements for this key. To manually update this policy, select **Edit**. Modifying the statement identifiers (Sid) assigned in the previous steps might affect how th updates to that statement.

```
1 {
2   "Id": "key-consolepolicy-3",
3   "Version": "2012-10-17",
4   "Statement": [
5     {
6       "Sid": "Enable IAM User Permissions",
7       "Effect": "Allow",
8       "Principal": {
9         "AWS": "arn:aws:iam::679625722057:root"
10      },
11      "Action": "kms:*",
12      "Resource": "*"
13    },
14    {
15      "Sid": "Allow access for Key Administrators",
16      "Effect": "Allow",
17      "Principal": {
18        "AWS": [
19          "arn:aws:iam::679625722057:user/Admin",
20          "arn:aws:iam::679625722057:user/ARJM"
21        ]
22      },
23      "Action": [
24        "kms:Create*",
25        "kms:Describe*",
26        "kms:Enable*",
```

Cancel

I have created one KMS (key management service)

☐	Aliases	Key ID	Status	Key type	Key spec	Key usage
☐	alias/h3-pii-key	97224261-3e1d-4d5a-8d8a-637b08a9f057	Enabled	Symmetric	SYMMETRIC_DEFAULT	Encrypt and decrypt

Enabled rotation and provided 180 days rotation time

KMS > Customer managed keys > Key ID: 97224261-3e1d-4d5a-8d8a-637b08a9f057 > Edit automatic key rotation

Key Management Service (KMS)

AWS managed keys
Customer managed keys

Custom key stores
AWS CloudHSM key stores
External key stores

Edit automatic key rotation

Automatic key rotation

Key rotation
☒ Enable
☐ Disable

Rotation period (in days)
Rotation period defines the number of days between each automatic rotation.
180
Enter a rotation period between 90 and 2560 days.

Cancel Save

Created one bucket disable public access

Bucket name

secure-pi-bucket-arg

Copy settings from existing bucket - optional

Only the bucket settings in the following configuration are copied.

Choose bucket

Format: <S3://bucket/prefix>

Object Ownership

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

ACLs disabled (recommended)

All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

ACLs enabled

Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

Object Ownership

Bucket owner enforced

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases.

Block all public access

Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

Block public access to buckets and objects granted through new access control lists (ACLs)

S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.

Block public access to buckets and objects granted through any access control lists (ACLs)

S3 will ignore all ACLs that grant public access to buckets and objects.

Block public access to buckets and objects granted through new public bucket or access point policies

S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

Block public and cross-account access to buckets and objects through any public bucket or access point policies

S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures.

Bucket Versioning

Disable

Enable

Tags - optional

You can use bucket tags to analyze, manage and specify permissions for a bucket.

You can use x3::TagForResource, x3::TagResource, and x3::UntagResource APIs to manage tags on S3 general purpose buckets for access control in addition to cost allocation and resource organization. To ensure a seamless transition, please provide permissions to x3::TagForResource, x3::TagResource, and x3::UntagResource actions.

No tags associated with this bucket.

Add new tag

You can add up to 50 tags.

Default encryption

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type

Secure your objects with two separate layers of encryption. For details on pricing, see SSE-KMS pricing on the Storage tab of the Amazon S3 pricing page.

Server-side encryption with Amazon S3 managed keys (SSE-S3)

Server-side encryption with AWS Key Management Service keys (SSE-KMS)

Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)

Bucket Key

Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS.

Disable

Enable

Advanced settings

After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.

Cancel Create bucket

Enabled encryption and provided key name

Default encryption

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type

Server-side encryption with Amazon S3 managed keys (SSE-S3)

Server-side encryption with AWS Key Management Service keys (SSE-KMS)

Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)

AWS KMS key

Choose from your AWS KMS keys

Enter AWS KMS key ARN

Available AWS KMS keys

arn:aws:kms:us-east-1:679625722057:key/97224261-3e1d-4d5a-8c8a-637b88a5f037

Create a KMS key

Changing the default encryption settings might cause in-progress replication and Batch Replication jobs to fail. These jobs might fail because of missing AWS KMS permissions on the IAM role that's specified in the replication configuration. If you change the default encryption settings, make sure that this IAM role has the necessary AWS KMS permissions.

Bucket Key

Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS.

Disable

Enable

Blocked encryption types

SSE-C (Server-Side Encryption with Customer-Provided Keys)

Upcoming change to default encryption

In April 2026, server-side encryption with customer-provided keys (SSE-C) will be blocked by default for all new buckets. If you need to use SSE-C encryption, make sure that SSE-C is not selected under Blocked encryption types.

Cancel Save changes

We can confirm that there is encryption role is attached to the bucket

The screenshot displays the Amazon S3 console interface for a bucket named 'secure-pil-bucket-arj'. The top navigation bar shows the bucket name and a file icon. Below the navigation bar, there are tabs for 'Properties', 'Permissions', and 'Versions'. The 'Permissions' tab is selected, showing the 'Access control list (ACL)'. A message states: 'This bucket has the bucket owner enforced setting applied for Object Ownership. When bucket owner enforced is applied, use bucket policies to control access.' Below this, a table lists the grantees and their permissions:

Grantee	Object	Object ACL
Object owner (your AWS account) Canonical ID: e5f7d9ba2bee423f1bb0c430d4323357d52b68196c8a495d0637776d0	Read	Read, Write
Everyone (public access) Group: http://acs.amazonaws.com/groups/global/AllUsers	-	-
Authenticated users group (anyone with an AWS account) Group: http://acs.amazonaws.com/groups/global/AuthenticatedUsers	-	-

Below the ACL table, the 'Server-side encryption settings' are shown. The 'Encryption type' is 'Server-side encryption with AWS Key Management Service keys (SSE-KMS)'. The 'Encryption key ARN' is 'arn:aws:kms:us-east-1:679625722057:key/97224261-3e1d-4d5a-8c8a-637b88a5f037'. The 'Bucket Key' is 'Enabled'.

KMS TASK 2-How do you encrypt EBS volumes for EC2 using a customer-managed KMS key and verify it?

✓ Steps to Follow

1. Create or reuse a customer-managed KMS key for EBS encryption.
2. Enable automatic key rotation on the KMS key.
3. Go to EC2 → Volumes → Create volume.
4. Enable Encryption and select the CMK.
5. Attach the encrypted volume to an EC2 instance.
6. Verify the volume shows Encrypted: Yes with the CMK.
7. Create an encrypted snapshot from the volume.

🔗 What this proves in real companies

- Disks are protected even if copied or stolen
- Meets security & compliance audits
- Enforces data-at-rest encryption standard

Created a volume and added encryption kms key

KMS key [info](#)

alias/s3-pii-key

KMS key description

KMS key owner

KMS key ID

KMS key ARN

Volumes that are created from encrypted snapshots are automatically encrypted using the same key as the snapshot, or using a different key that you specify. Volumes that are created from unencrypted snapshots are automatically unencrypted, but you can choose to encrypt them using a specific key. If no snapshot is selected, you can choose to encrypt the volume and specify your own key. [Learn More](#)

Tags - optional [info](#)

Attach that volume to the ec2

EC2 > Volumes > vol-086c889e596626dc8 > Attach volume

Attach volume [info](#)

Attach a volume to an instance to use it as you would a regular physical hard disk drive.

Basic details

Volume ID

vol-086c889e596626dc8

Availability Zone

use1-az1 (us-east-1a)

Instance [info](#)

i-Ocft19db2428489e5

Device name [info](#)

/dev/sdb

Cancel Attach volume

We can see volume is successfully attached to the KMS key

Encryption Encrypted	KMS key ID 97224261-3e1d-4d5a-8c8a-637b88a5f037	KMS key alias alias/s3-pii-key	KMS key ARN arn:aws:kms:us-east-1:679625722057:key/97224261-3e1d-4d5a-8c8a-637b88a5f037
Status checks	Monitoring	Tags	

Created snapshot using the secured volume which I have added to kms key

EC2 > Volumes > vol-086c889e596626dc8 > Create snapshot

Create snapshot [info](#)

Create a point-in-time snapshot to back up the data on an Amazon EBS volume to Amazon S3.

Source volume

Volume ID

vol-086c889e596626dc8

Availability Zone

use1-az1 (us-east-1a)

Snapshot details

Description

encryption volume

Tags

Add tag

Cancel Create snapshot

Snapshot also attached to the kms key

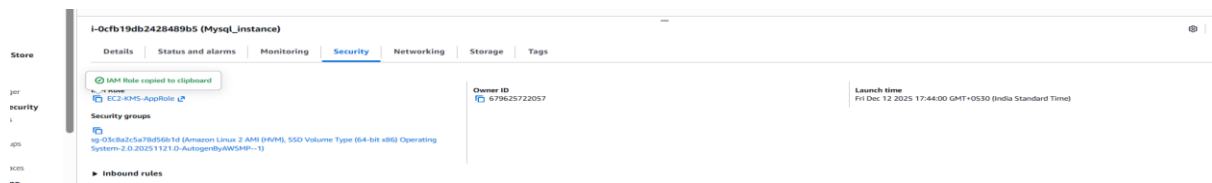
snap-00bb8d738524fa6d5				Last updated less than a minute ago		Delete	Actions
Details		Full snapshot size		Progress		Snapshot status	
Snapshot ID		0 B		100%		Completed	
Owner		Started		Product codes		Fast snapshot restore	
679625722057		Thu Dec 18 2025 14:51:55 GMT+0530 (India Standard Time)		-		-	
Description		Volume size					
encryption volume		10 GiB					
Source volume							
Volume ID							
vol-086c889e596626dc8							
Encryption		KMS key ID		KMS key alias		KMS key ARN	
Encrypted		97224261-3e1d-4d5a-8c8a-637b88a5f037		alias/s3-pii-key		arn:aws:kms:us-east-1:679625722057:key/97224261-3e1d-4d5a-8c8a-637b88a5f037	

KMS TASK 3 — How do you restrict a KMS key so that only a specific IAM role can decrypt data?

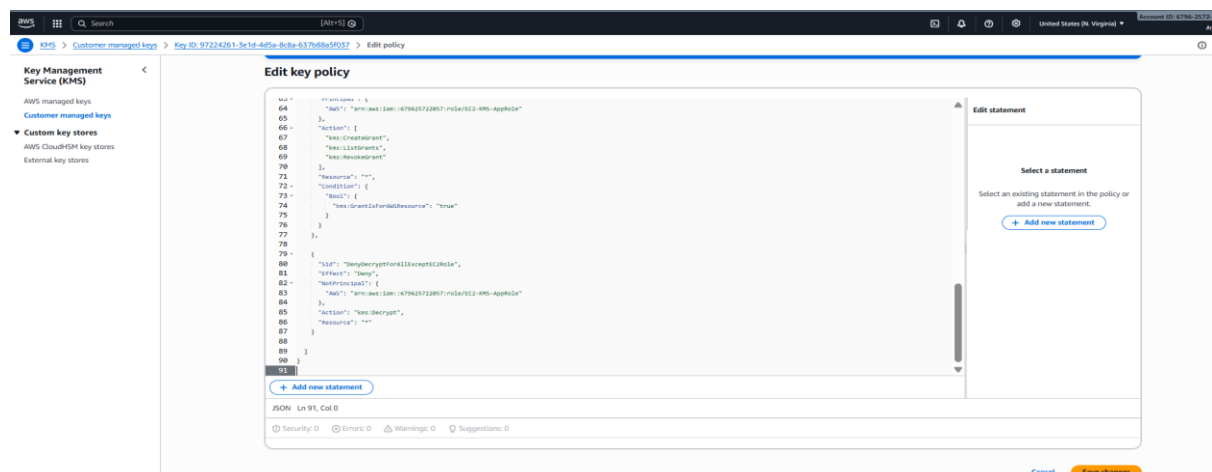
✓ Steps to Follow (ONLY 7 Lines)

1. Identify the application IAM role (EC2 or Lambda role).
2. Open AWS KMS → Customer-managed keys → select key.
3. Go to Key policy → Edit (JSON view).
4. Allow kms:Decrypt, Encrypt, GenerateDataKey to the app role.
5. Add an explicit Deny for all other principals.
6. Save the key policy and test access using the app role.
7. Confirm unauthorized users get AccessDenied.

Ec2 IAM role



1. Open AWS KMS → Customer-managed keys → select key.
2. Go to Key policy → Edit (JSON view).
3. Allow kms:Decrypt, Encrypt, GenerateDataKey to the app role.
4. Add an explicit Deny for all other principals.
5. Save the key policy and test access using the app role.



SECRETS MANAGER – How do you securely store database credentials using AWS Secrets Manager and allow an application to retrieve them?

✓ Steps to Follow

1. Create a customer-managed KMS key for Secrets Manager.
2. Create a secret storing DB username and password.
3. Encrypt the secret using the KMS CMK.
4. Name the secret using a standard path format.
5. Attach a Secrets Manager read policy to the app role.
6. Retrieve the secret from EC2 or Lambda.
7. Verify access using CloudTrail logs.

AWS Secrets Manager
Security, Identity, & Compliance

AWS Secrets Manager
Easily rotate, manage, and retrieve secrets throughout their lifecycle

AWS Secrets Manager helps you protect access to your applications, services, and IT resources. You can easily rotate, manage, and retrieve database credentials, API keys, and other secrets throughout their lifecycle.

Get started
You can store database credentials or any other type of secret.
[Store a new secret](#)

Pricing
\$0.40 per secret per month
\$0.05 per 10,000 API calls
[Learn more](#)

How it works
Use Secrets Manager to store, rotate, monitor, and control access to secrets such as database credentials, API keys, and OAuth tokens. Enable secret rotation using built-in integration for MySQL, PostgreSQL, and Amazon Aurora on Amazon RDS. You can also turn on rotation for other secrets using [Secrets Manager Rotation](#). To maintain control over your secrets, Secrets Manager can be used to rotate secrets for other services.

Secret type [Info](#)

☒ Credentials for Amazon RDS database ☐ Credentials for Amazon DocumentDB database ☐ Credentials for Amazon Redshift data warehouse

☐ Credentials for other database ☐ Managed external secret
Secrets vended by your third-party software vendor. ☐ Other type of secret
API key, OAuth token, other.

Credentials [Info](#)

User name
dbadmin

Password

☐ Show password

Encryption key [Info](#)
You can encrypt using the KMS key that Secrets Manager creates or a customer managed KMS key that you create.
aws/secretsmanager [Add new key](#)

Database [Info](#)

DB instance	DB engine	Status	Creation date (UTC)
☒ database-1	mariadb	available	December 11, 2025 at 14:48:33

[Cancel](#) [Next](#)

Resource permissions - optional [info](#)

Cancel

Save

Add or edit a resource policy to access secrets across AWS accounts.

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": "secretsmanager:GetSecretValue",
7       "Resource": "arn:aws:secretsmanager:*:679625722057:secret:prod/rds/mysql/*"
8     }
9   ]
10 }
11
```

prod/rds/mysql/credentials

🕒 You successfully stored the secret prod/rds/mysql/credentials. To show it in the list, choose Refresh. AWS CloudFormation is setting up rotation resources, this can take up to 2 minutes to complete.

Use the sample code to update your applications to retrieve this secret.

[View details](#) [See sample code](#)

prod/rds/mysql/credentials

Secret details

Encryption key

aws/secretsmanager

Secret name

prod/rds/mysql/credentials

Secret ARN

arn:aws:secretsmanager:us-east-1:679625722057:secret:prod/rds/mysql/credentials-xOcZfm

Secret description

Production MySQL database credentials

Secret type

Overview

Rotation

Versions

Replication

Tags

Secret value

info

Retrieve and view the secret value.

Resource permissions

- optional

info

Add or edit a resource policy to access secrets across AWS accounts.

Sample code

Use these code samples to retrieve the secret in your application.

Java

JavaScript

C#

Python3

Ruby

Go

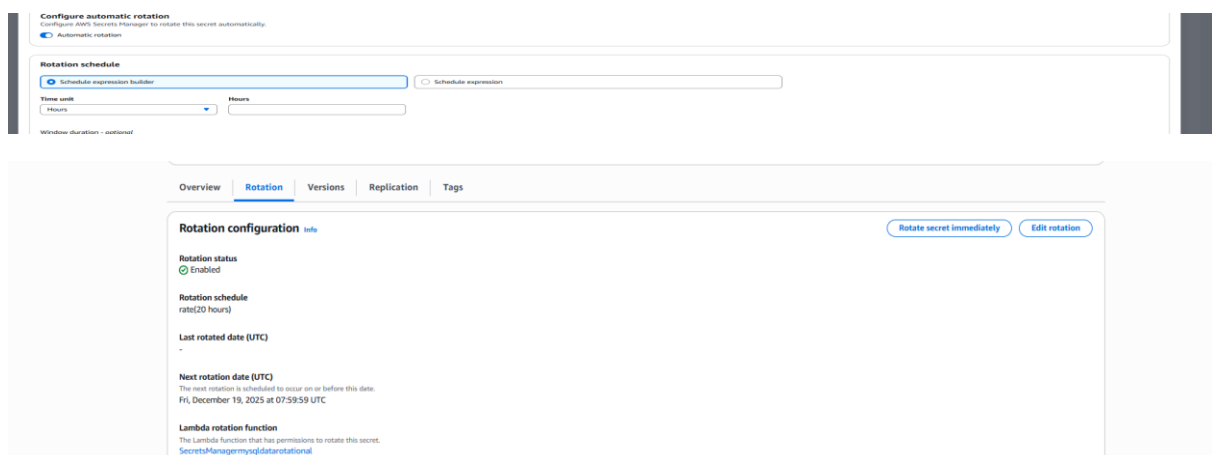
Rust

PHP

SECRETS MANAGER TASK 2 —How do you enable automatic rotation of database credentials in AWS Secrets Manager?

✓ Steps to Follow

1. Open the existing database secret in Secrets Manager.
2. Enable automatic rotation.
3. Choose AWS-managed rotation Lambda for RDS.
4. Attach the Lambda to the secret.
5. Set rotation interval to 30 days.
6. Run Rotate secret now.



SECRETS MANAGER TASK 3-- How do you centrally manage third-party API keys using AWS Secrets Manager and allow applications to access them securely?"

✓ Steps to Follow (Short – 7 Lines)

1. Create a secret to store third-party API keys.
2. Encrypt the secret using a KMS CMK.
3. Grant read-only access to application IAM roles.
4. Retrieve API keys dynamically from the app.
5. Rotate API keys by updating the secret.

6. Verify apps use the new key without redeploy.

7. Audit secret access using CloudTrail.

prod/api/thirdparty/keys

Secret details

Encryption key
aws/secretsmanager

Secret name
prod/api/thirdparty/keys

Secret ARN
arn:aws:secretsmanager:us-east-1:679625722057:secret:prod/api/thirdparty/keys-e4HJas

Secret description
Production third-party API keys

Secret type
-

Overview

Rotation

Versions

Replication

Tags

Secret value Info
Retrieve and view the secret value.

Close Edit

```
[ec2-user@ip-12-0-0-167 ~]$ aws secretsmanager get-secret-value \
--secret-id prod/api/thirdparty/keys
{
  "ARN": "arn:aws:secretsmanager:us-east-1:679625722057:secret:prod/api/thirdparty/keys-e4HJas",
  "Name": "prod/api/thirdparty/keys",
  "VersionId": "c0b782f7-d3d5-4116-93ec-7cfe85687b77",
  "SecretString": "{\"TWILIO_API_KEY\":\"sk_live_XXXXX\", \"STRIPE_API_KEY\":\"sk_test_XXXXX\", \"SENDGRID_API_KEY\":\"SG.XXXXXX\"}",
  "VersionStages": [
    "AMSCURRENT"
  ],
  "CreatedDate": "2025-12-18T11:57:12.900000+00:00"
}
```

Overview

Rotation

Versions

Replication

Tags

Secret value Info
Retrieve and view the secret value.

Close Edit

Key/value

Plaintext

Secret key	Secret value
TWILIO_API_KEY	sk_live_XXXXX
STRIPE_API_KEY	sk_test_XXXXX
SENDGRID_API_KEY	SG.XXXXXX

Resource permissions - optional Info
Add or edit a resource policy to access secrets across AWS accounts.

Edit permissions

```
[ec2-user@ip-12-0-0-167 ~]$ aws secretsmanager get-secret-value --secret-id prod/api/thirdparty/keys
{
  "ARN": "arn:aws:secretsmanager:us-east-1:679625722057:secret:prod/api/thirdparty/keys-e4HJas",
  "Name": "prod/api/thirdparty/keys",
  "VersionId": "c01de36a-d11a-4bb4-838e-66a6c1621e3a",
  "SecretString": "{\"TWILIO_API_KEY\":\"sk_live_arj\", \"STRIPE_API_KEY\":\"sk_test_XXXXX\", \"SENDGRID_API_KEY\":\"SG.XXXXXX\"}",
  "VersionStages": [
    "AMSCURRENT"
  ],
  "CreatedDate": "2025-12-18T12:06:41.189000+00:00"
}
```

Retrieved the old values

```
[ec2-user@ip-12-0-0-167 ~]$ aws secretsmanager get-secret-value --secret-id prod/api/thirdparty/keys
{
  "ARN": "arn:aws:secretsmanager:us-east-1:679625722057:secret:prod/api/thirdparty/keys-e4HJas",
  "Name": "prod/api/thirdparty/keys",
  "VersionId": "c01de36a-d11a-4bb4-838e-66a6c1621e3a",
  "SecretString": "{\"TWILIO_API_KEY\":\"sk_live_arj\", \"STRIPE_API_KEY\":\"sk_test_XXXXX\", \"SENDGRID_API_KEY\":\"SG.XXXXXX\"}",
  "VersionStages": [
    "AMSCURRENT"
  ],
  "CreatedDate": "2025-12-18T12:06:41.189000+00:00"
}
[ec2-user@ip-12-0-0-167 ~]$ aws secretsmanager get-secret-value \
--secret-id prod/api/thirdparty/keys \
--version-stage AMSPREVIOUS
{
  "ARN": "arn:aws:secretsmanager:us-east-1:679625722057:secret:prod/api/thirdparty/keys-e4HJas",
  "Name": "prod/api/thirdparty/keys",
  "VersionId": "c0b782f7-d3d5-4116-93ec-7cfe85687b77",
  "SecretString": "{\"TWILIO_API_KEY\":\"sk_live_XXXXX\", \"STRIPE_API_KEY\":\"sk_test_XXXXX\", \"SENDGRID_API_KEY\":\"SG.XXXXXX\"}",
  "VersionStages": [
    "AMSPREVIOUS"
  ],
  "CreatedDate": "2025-12-18T11:57:12.900000+00:00"
}
```